Lab: Standard Normal Distribution

Estimated reading time: 30 minutes

In this lab, you will learn how to find the standard normal distribution using Microsoft Excel.

Software used in this lab

The instruction videos in this course use the full Excel Desktop version as this has all the available product features, but for the hands-on labs, we will be using the free "Excel for the web" version as this is available to everyone.

Although you can use the Excel Desktop software if you have access to this version, it is recommended that you use Excel for the web for the hands-on labs, as the lab instructions specifically refer to this version, and there are some minor differences in the interface and available features.

Learning Objectives:

- Calculate the standard normal distribution for a given value when the mean and standard deviation are specified.

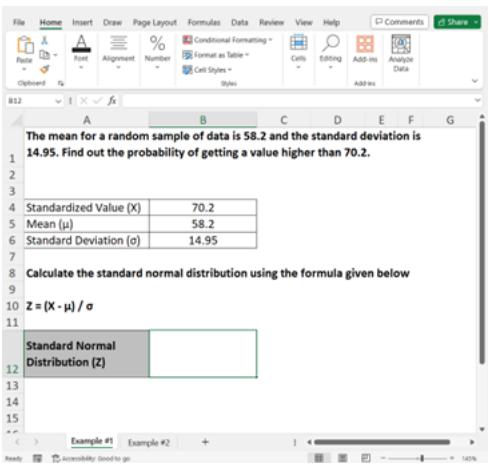
Welcome to the Lab: Standard Normal Distribution!

Download the file Standard-Normal-Distribution_problems.xlsx. To download the file, right-click and open the link in a new tab.

Follow the steps below to solve the problems.

Exercise 1: Calculate the standard normal distribution using the formula

The mean for a random sample of data is 58.2, and the standard deviation is 14.95. Determine the corresponding standard normal distribution (Z-score) for a value of 70.2.



X is the value for which you want to find the distribution.

Standardized value (X) = 70.2

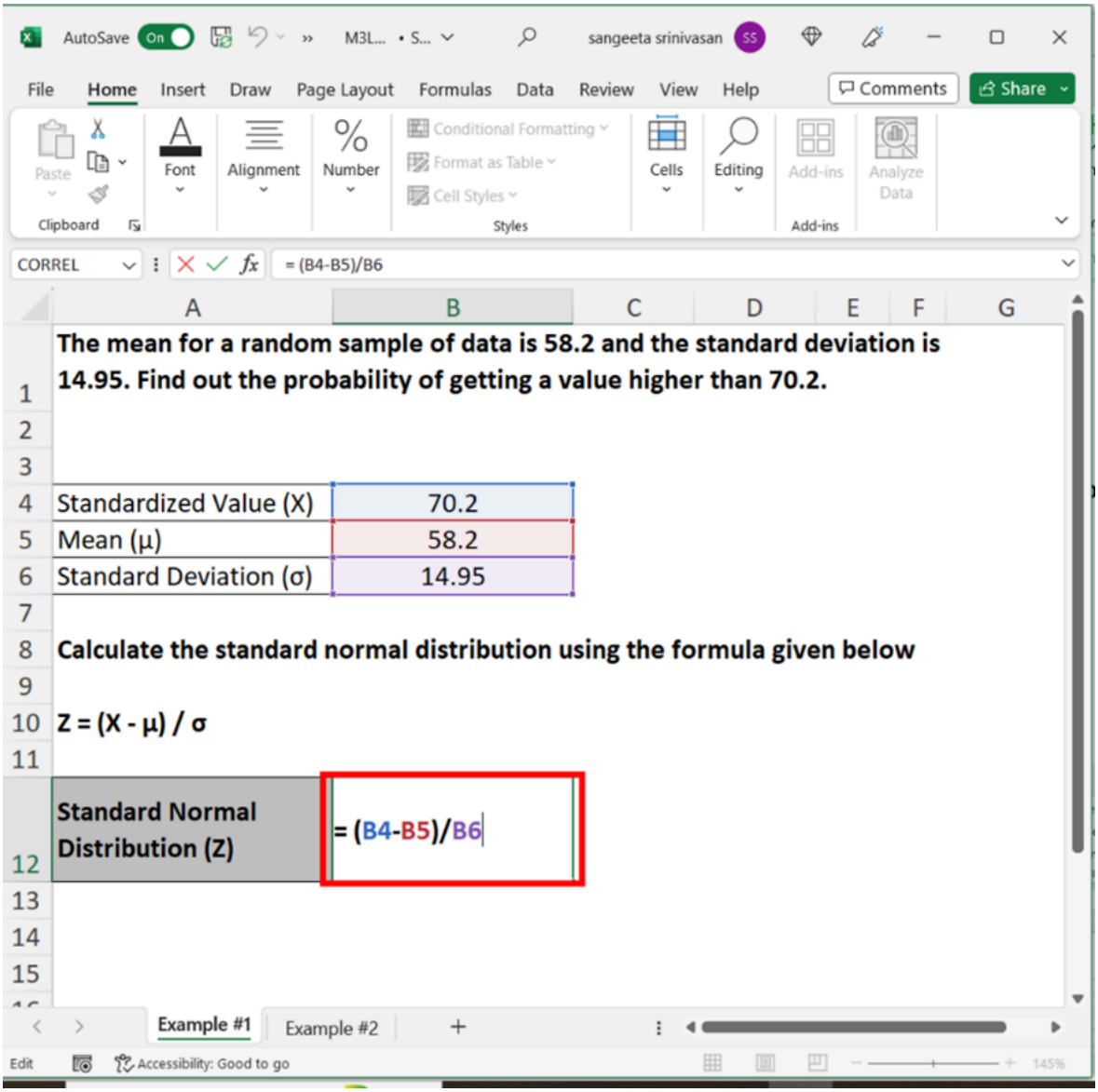
Mean (μ) = 58.2

Standard deviation (σ) = 14.95

The standard normal distribution (z) is calculated as given below:

Z = (X – μ) / σ

Perform the calculations in Excel as shown in the below screenshot.



The standard normal distribution for X value of 70.2 is equal to **0.803**.

Exercise 2: Calculate the standard normal distribution using Excel

An electric-bike travels at a top speed of 120 km per hour, whereas the minimum speed is 30 km per hour. Thus, the average speed at which the electric bike travels is 80 km per hour. If the standard deviation is 9, determine the corresponding standard normal distribution (Z-score) for a speed of 90 km per hour.

X is the value for which you want to find the distribution.

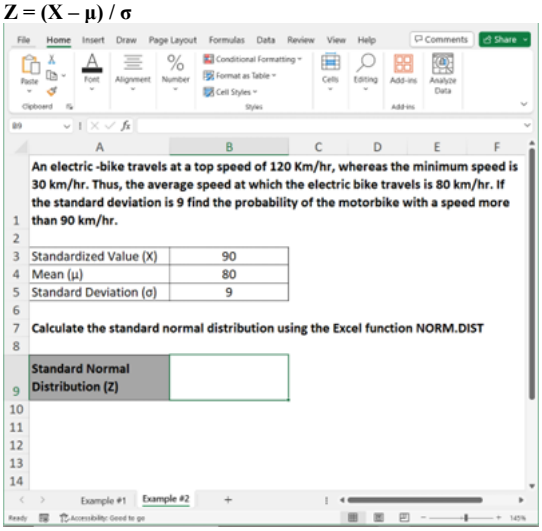
Standardized value (X) = 90

Mean (μ) = 80

Standard deviation (σ) = 9

The standard normal distribution (z) is calculated as given below:

$Z = (X - \mu) / \sigma$



The standard normal distribution for X value of 90 km per hour is equal to **1.11**.

Congratulations! You've completed the lab activity on standard normal distribution, and you are ready to move to the next topic.

Author(s)

- Dr. Rajendra Patil

Other Other Contributor(s)

- Rashi Kapoor
- Sangeeta Srinivasan